Forecasting Cryptocurrency Price Trends Using ARIMA and LSTM

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ABSTRACT

The Bitcoin prices will change rapidly due to several factors, such as market demand, investor activity, and news articles, making it a very difficult task to predict them. The purpose of this study is to predict the short-term bitcoin price changes that can be done with statistical and deep learning methods, which are ARIMA and Long Short-Term Memory(LSTM) models. The Historical price data of 1-minute duration is gathered and preprocessed, and Exploratory data analysis suggests the non-stationary properties of the dataset, which are covered with the help of differencing techniques. ARIMA model is used to predict linear trends, whereas the LSTM model is trained to identify non-linear and very complicated patterns and trends, and the models are evaluated using by Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and Mean Absolute Percentage Error (MAPE). The findings indicate that although ARIMA represents general tendencies, LSTM is far more effective compared to ARIMA because it is able to reproduce the general trends as well as the volatility of the Bitcoin prices.

I. INTRODUCTION / BACKGROUND

The cryptocurrency has become one of the most important elements of the financial environment, and Bitcoin has become the biggest digital asset. This market will run 24/7 and influenced by financial markets that are susceptible to investor sentiment, economic factors, and technology, which determine the cryptocurrency market. This promotes severe instability in the price, so it is crucial and difficult to predict [4].

Proper prediction of cryptocurrency prices is important in supporting informed decision-making by investors, traders, and financial analysts. Time-series forecasting with the help of traditional statistical models, such as ARIMA, has gained popularity because it allows for modeling linear relationships.These models do not normally exhibit the ability to model the non-linear and complex behavior of financial data.

To address these limitations, deep learning models like Long Short-Term Memory (LSTM) networks have become popular. Long-term dependencies and non-linear patterns of sequential data can be learned by LSTM. This paper will integrate both ARIMA and LSTM to compare the effectiveness of forecasting price fluctuations of Bitcoin as well as the most appropriate model to be used in short-term forecasting [2].

II. PROBLEM STATEMENT

It is natural that the market is very volatile and nonlinear because the prediction of cryptocurrency prices is complicated. The various factors that affect price movements are varied and include the actions of the investors, the financial activities across the globe, technological advancements, and the social media trends. The above factors present unpredictability and randomness in the market, and it is therefore a challenging task to predict what will happen in the market at the right time. Conventional financial projection tools tend to assume stability and linearity, which are not valid in the cryptocurrency markets. Therefore, these procedures are ineffective in terms of pricing variability, which should be dynamic and time-dependent. Moreover, crypto data analysis is subjective and inefficient, as well as not very practical because of the high data volume and frequency [3].

A second significant problem is the existence of non-stationarity in time-series data, which is a statistical property/quantity in which the means and variance of statistical quantities vary over time. This renders the implementation of standard modeling functions hard unless data are transformed appropriately. As such, it is required to have an efficient and automated prediction system that is able to manage time-dependent data, capture underlying trends and are able to give accurate predictions. This project will satisfy this requirement by introducing the ARIMA model that is specifically intended to predict a time series, and that can successfully model the trends and dependencies of cryptocurrency price data [5].

III. LITERATURE REVIEW / RELATED WORK

The prediction of cryptocurrency prices by using statistical and machine learning methods has received extensive research. The article by Sailaja et al. (2024) investigated ARIMA and Facebook Prophet to identify Ethereum tendencies and proved that statistical models may be effective in capturing causal dynamics of financial information. Their research study emphasized the fact that time-series analysis is an important tool to model cryptocurrency markets.

They used the ARIMA model to predict Bitcoin prices and demonstrated that it is useful in short-term forecasting as it incorporates time-related information in historical data. On the same note, performed a comparative inference accof ARIMA and regression model, and found out that the performance of ARIMA in the model sequential data is better because of its capability to address time-conscious dependencies [1].

They considered the conventional ARIMA models and combined them with GARCH models to consider financial data volatility. This combination method enhanced the precision of prediction in that it modeled the mean and variance of time-series data. They also showed that ARIMA successfully predicts the price of Bitcoin and is simple and easy to understand as opposed to more complicated models.

Moreover, suggested hybrid methods according to which statistical models are used to combine machine learning approaches to improve the accuracy of predictions. Their results indicate that although superior models can be used to obtain better results, ARIMA is also a formidable and consistent time-series prediction model.

All these studies point to the evidence that ARIMA is a proven and efficient technique of price forecasting of cryptocurrencies, and it is an appropriate tool in this project [6].

IV. METHODOLOGY / PROPOSED APPROACH

The plan will be based on a systematic pipeline comprising of statistical and deep learning strategies to forecast Bitcoin prices precisely. A 60-day timeframe is selected to collect historical price data that is provided by the Binance API with a 1-minute time interval to obtain high-frequency data that can be used in the short term to predict. The collected dataset is preprocessed by transforming timestamps into a valid datetime format, treating missing data, cleaning, and standardizing all numerical data to be analyzed.

Exploratory data analysis helps to understand the performance of the data. Price trend visualization shows that the data is very volatile and uneven, which demonstrates the complexity of the data. Augmented Dickey-Fuller (ADF) test is conducted to assess the stationarity to be prepared to model the data. It has an initial p-value around 0.008, which means that the data is stationary. Still it is followed by first-order differencing to further stabilize the variance and eliminate minor fluctuations, and the p-value is significantly reduced to almost 0.00, which reinforces stationarity [7].

Application of differentiation is done not just to meet the assumptions in the model, but to enhance the performance of the model by eliminating noise and making patterns more consistent. which helps the ARIMA model to capture the trends and improves the learning ability of LSTM.

Based on the ACF (Autocorrelation function) and the PACF (Partial autocorrelation function), the initial selected ARIMA(1,1,0) [8].

An 80:20 split of the data is then done into training and testing sets. The ARIMA model is trained with the stationary data to linear dependencies. Simultaneously, the normalized data LSTM model is generated with MinMax scaling, and sequences generated with a sliding window method. The LSTM model is made up of several layers and one dense output layer, which is trained with the Adam optimizer and MSE loss.

Measures applied to model performance include Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) and the Mean, Absolute Percentage Error (MAPE). The evaluation is followed by the creation of the two models to make predictions within the next 60 minutes, allowing the assessment of how the two models forecast in the short term [9].

V. DATA ANALYSIS

A. Data Collection & Cleaning

The Bitcoin price data was collected by using the Binance API, where the dataset is having an 1 minute interval with 60 days period and has around 80000 records. The collected data was in BTC/USD. The dataset has features like open, high, low, Low, Close and value. The raw data is preprocessed by converting Unix timestamps into date and time formats, and the features like OHLC and volume are converted to appropriate data types and checked for missing and duplicate values. The columns that are not required are removed, and finally, the data is ready for analysis and modeling.

B. Exploratory Data Analysis (EDA)

Fig. 1. Bitcoin Close Price Over Time (1-minute intervals)

The Exploratory data analysis of Bitcoin's close price over time and Rolling mean and standard deviation plots helps to identify whether the data is stable or not.
The Bitcoin close price over time plot indicates the prices are fluctuating consistently with up and down spikes, which shows there is no constant mean level price, which indicates data is non-stationary.

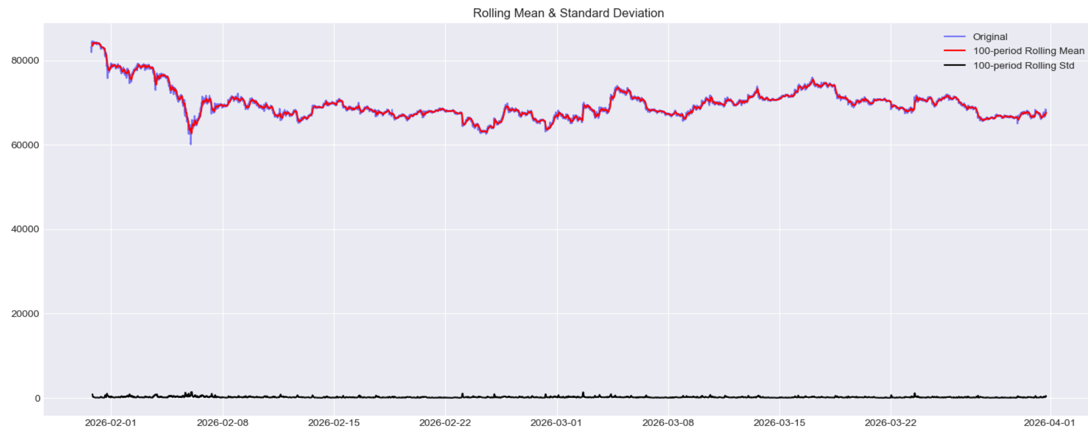

Fig. 2. Rolling Mean and Standard Deviation

The rolling mean shows continuous variations, which indicates the trend is changing over time, and the rolling standard deviation also fluctuates, which means the data is not stable. Since both mean and variance are changing, that confirms the time series is non-stationary, which is not suitable for the ARIMA model.

C. Outlier Analysis

The bitcoin close prices show some outliers, which indicate high market volatility. By using the IQR method, the outliers falling below 61969.20 and above 76208.32 were identified as outliers. Since 6121 outliers are detected, resulting in continuous sudden fluctuations, the outliers show the real market behavior influenced by some factors, such as news articles and investor activity. These outliers suggest that the data is highly dynamic.

D. Stationarity & ADF Test

The Exploratory Data Analysis of rolling mean and standard deviation shows the data is non-stationary. To check whether the data is stationary or not, statically i used the Augmented Dickey-Fuller test, which is a statistical test used to check whether the data is stationary or not. Initially, the p-values were less then 0.05, but to eliminate minor fluctuations shown by EDA, I applied first-order differencing, which resulted in the p-value approximately 0.00, which indicates the data is strongly stationary, which is suitable for the ARIMA model

E. ACF & PACF Analysis

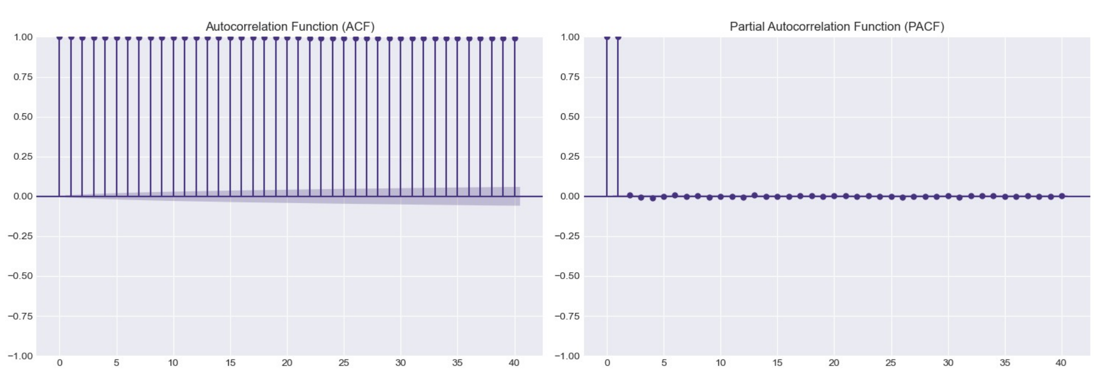

Fig. 3. ACF and PACF Plots

The ACF and PACF plots are used to identify the initial ARIMA parameters, where ACF indicates how previous values influence the current values. The ACF plot shows the strong coloration between current and past values it shows some slow decay but no sharp cuts, which shows there is no moving average component so the value q is set to 0, and the first order differencing is applied, d=1, and PACF, which indicates how previous values are directly influencing the current value, has a spike at lag 1 and drops to zero, so Autoregressive component p is set to 1. So based on ACF and PACF, the initial ARIMA(p,d,q)=(1,1,0)

F. Model Development

For the ARIMA Model, the data was converted into a stationary series by using first-order differencing to stabilize the mean. Initially, based on the ACF and PACF analysis graphs, initial model was suggested as ARIMA(1,1,0). The model was trained on the training dataset, which can learn the patterns. However, during model tuning and validation, and based on lower AIC and BIC values and lower RMSE.Therefore, the ARIMA(0,1,0) is selected as the final model because of its stable performance and reduced risk of overfitting.

For the LSTM model, the data was normalized using MinMax scaling of range 0-1, which can improve the stability. Then, the time series data was transformed into a sequence using a sliding window with an approach of 60 time steps, which helps the model to learn dependencies over fixed intervals The LSTM model consists of 2 layers of 50 units each with 0.2 dropout to reduce the overfitting, and with a dense output layer during model training, both training and validation losses are decreased consistently and stabilize over epochs from the graph. The validation loss and training loss are closer to each other, which indicates the model is avoiding overfitting and showing some strong generalization over unseen data.

G. Actual vs Predicted Comparison

Fig. 4. Actual vs Predicted

The above graph shows the comparison graphs between actual and predicted values between ARIMA and LSTM models.

From the graph, the ARIMA model produces stable and smooth predictions. The model is capturing the overall trend but showing a limited response towards some sudden price changes. The flat line occurs in the graph because the selected ARIMA model(0,1,0), which indicates the model is not having the autoregression component and moving average component, only relies on differencing d=1, which shows the predictions are based on the last observed values.

The LSTM model follows the Actual predictions closely, which clearly shows the model is capturing both trend and sudden price changes, indicating the model has the ability to learn nonlinear relations, and also the model adapts to sudden changes.

H. Residual Analysis

Fig. 5. ARIMA and LSTM Residuals and Distribution

I. Residual Analysis

The purpose of residual analysis is to understand the prediction errors of the models. The Residuals of ARIMA show some clear patterns and large variation, which indicates that the model fails to capture some important changes in the data.

The Residuals of the LSTM model appear with lower variance and are randomly distributed around zero, which indicates the model is capturing both linear and nonlinear patterns. Therefore, LSTM demonstrates a better overall fit compared to the ARIMA.

J. Forecasting Behavior (Next 60 Minutes)

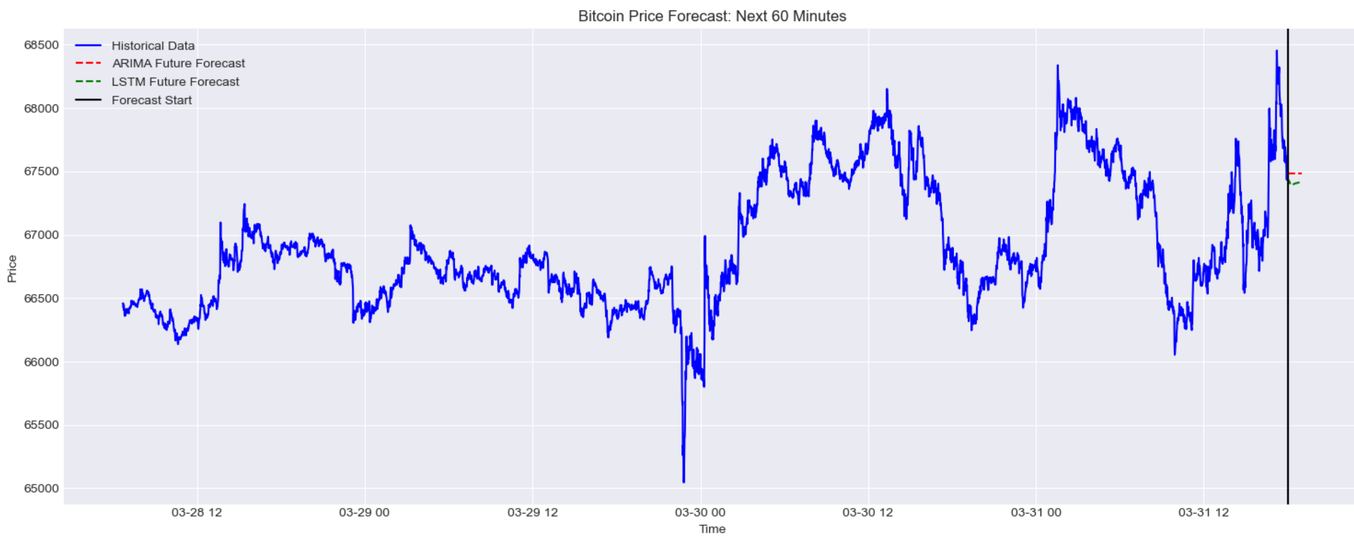

Fig. 6. Bitcoin Price Forecast (Next 60 Minutes – ARIMA vs LSTM)

The forecasting results of both models for the next 60 minutes show how the models will behave in the future. The ARIMA model is gives stable predictions that follows overall trend but show limited response towards sudden price changes because ARIMA model is based on linear relations. On the other side the LSTM model is capturing sudden changes, which is adapting to trends and changes. Therefore, LSTM is more suitable for real-time forecasting in dynamic markets.

K. Model Evaluation

The model performances are evaluated using evaluation metrics such as Mean Absolute Error(MAE), Root Mean Square Error (RMSE), and Mean Absolute Percentage Error (MAPE). These measures are ways of measuring accuracy in predictions and scores of actual and predicted scores.

The findings indicate that there is a significant difference in performance between the two models. The ARIMA model gives more forecasts and does not capture the sudden price changes, leading to high error terms. Instead, the LSTM model adapts to real price fluctuations and reflects the temporary fluctuations as well as long-term tendencies.

TABLE I
MODEL PERFORMANCE COMPARISON

Model	MAE	RMSE	MAPE
ARIMA	466.92	577.49	0.70%
LSTM	64.41	94.64	0.09%

These findings can further be supported by residual analysis, where ARIMA residual plots depict discernible patterns with greater variance, whereas LSTM residual plots are more clumped around zero, which is a superior fit of any model. In general, LSTM would be better at predicting volatile cryptocurrency data compared to ARIMA due to its better predictive value and reliability.

VI. RESULTS AND DISCUSSIONS

These findings are a clear indication that there is a clear variation in the results of the ARIMA and LSTM models used in the forecasting of the Bitcoin price. According to the assessment measures, the LSTM model has significantly low error values as opposed to ARIMA, a fact that proves its high predictative behavior.

The overall trend of the Bitcoin price series can be effectively captured using ARIMA model. It generates smooth and stable forecasts which are directed ,by the general trend of data. But being based on linear assumptions, ARIMA is not responsive enough to any sudden shifts in the market and sudden price changes. This restriction renders it inapplicable to very volatilized financial information like cryptocurrency prices.

Preferably, the LSTM model demonstrates a significantly high competence in capturing both short-term and long-term dependencies. It dynamically adapts to the changes in the data, and produces predictions that closely correspond to the actual price movements. This is especially relevant in cryptocurrency markets, where non-linear influences in price behavior drive the prices.

This residual analysis confirms that the LSTM model is the best. The ARIMA model residuals portray observable trends and are largely scattered around, which shows that the model has not successfully inherent construction of the information. Conversely, the LSTM model residuals are distributed randomly, and this is an indication that a large number of the patterns in the data have been well learnt.

Besides, future forecasts in the next 60 minutes show the usefulness of LSTM. Although ARIMA results in a smoother forecast (not responsive), LSTM forecasts are adaptive and capture the recent trends and volatility. This renders LSTM more dependable to make short-term decisions regarding dynamic markets.

On the whole, the experimental findings support the idea that although ARIMA may be used as a benchmark to make trend analysis, LSTM is much more successful in terms of accuracy, flexibility and stability [10].

VII. CONCLUSION

Finally, in this project, I have implemented two effective models, which are statistical and deep learning models, to predict bitcoin prices. The ARIMA model is capturing the overall trends effectively, but the model is struggling to capture the sudden changes

Another approach, the LSTM model, shows strong performance in capturing both linear and non-linear patterns, and the model is able to adapt to sudden changes and gives better predictions. The reduced error measures and the enhanced residual characteristics highlight the LSTM model.

Overall, from this analysis the LSTM model is more suitable for forecasting cryptocurrency prices because of their ability to handle complex and volatile data patters

To improve the model, the target of future work could be to add more features to the structures trading volume, market sentiment, and macroeconomic indicators. It is also possible, to experiment with hybrid models that are a combination of ARIMA and deep learning methods to use the advantages of both methods and achieve a high level of accuracy

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